This CVPR paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

# F2Net: A Frequency-Fused Network for Ultra-High Resolution Remote Sensing Segmentation

Hengzhi Chen[1] Liqian Feng[1] Wenhua Wu[1] Xiaogang Zhu[2,*]
Qiuxia Wu[3] Lianlei Shan[4] Kun Hu[5,*]
[1]The University of Sydney [2]Adelaide University [3]South China University of Technology
[4]Tsinghua University [5]Edith Cowan University
{hche6958,lfen0902,wewu6176}@uni.sydney.edu.au, xiaogang.zhu@adelaide.edu.au,
qxwu@scut.edu.cn, shanglianlei18@mails.ucas.edu.cn, k.hu@ecu.edu.au

## Abstract

*Semantic segmentation of ultra-high-resolution (UHR) remote sensing imagery is critical for applications like environmental monitoring and urban planning but faces computational and optimization challenges. Conventional methods either lose fine details through downsampling or fragment global context via patch processing. While multi-branch networks address this trade-off, they suffer from computational inefficiency and conflicting gradient dynamics during training. We propose F2Net, a frequency-aware framework that decomposes UHR images into high- and low-frequency components for specialized processing. The high-frequency branch preserves full-resolution structural details, while the low-frequency branch processes downsampled inputs through dual sub-branches capturing short- and long-range dependencies. A Hybrid-Frequency Fusion module integrates these observations, guided by two novel objectives: Cross-Frequency Alignment Loss ensures semantic consistency between frequency components, and Cross-Frequency Balance Loss regulates gradient magnitudes across branches to stabilize training. Evaluated on DeepGlobe and Inria Aerial benchmarks, F2Net achieves state-of-the-art performance with mIoU of 80.22 and 83.39, respectively.*[1]

## 1. Introduction

Semantic segmentation of remote sensing (RS) imagery is fundamental to various applications, including geospatial analysis [32], urban planning [33], and environmental monitoring [18]. With advances in RS technologies, ultra-high-resolution (UHR) imagery has become increasingly accessible, offering unprecedented spatial detail. While $2048 \times 1080$ images are conventionally considered high resolution [1], UHR typically refers to those exceeding 4K resolution [41]. For segmentation tasks, UHR images provide rich spatial context, but also introduce substantial computational challenges due to their size and detail density.

Conventional segmentation methods are tailored for moderate-resolution inputs (e.g., $256 \times 256$), which are $\sim 100\times$ smaller than UHR images. To adapt, early approaches either downsample the full image [2, 45], sacrificing fine details, or adopt patch-wise processing [30], which preserves local structure but loses global context. Recent advances move toward multi-branch architectures that jointly model local detail and global semantics. GLNet [5] introduces dual branches for coarse global input and high-resolution patches. ISDNet [12] improves this by directly extracting fine details from UHR inputs without cropping. SGNet [34] further generalizes this paradigm by leveraging surrounding context to enhance patch-level reasoning. Despite improved performance, these methods face two key issues: (1) high computational cost due to redundant inference (e.g., GLNet requires $\sim$200 patches for a single UHR image); and (2) conflicting gradients across branches during joint training, leading to optimization imbalance and misaligned features.

Therefore, we propose F2Net - a frequency-aware network. By decomposing UHR imagery into high- and low-frequency components, F2Net comprises three frequency aware branches: a high-frequency branch for detail perservation and two low-frequency subbranches for semantic extraction. The high-frequency branch retains full-resolution inputs and focuses on capturing finer details, such as edges and textures, which are critical for accurate boundaries. In contrast, the low-frequency branch processes a downsampled version of the image, encoding semantically rich yet spatially redundant features. This low-frequency branch is further divided into two sub-branches: one modeling short-

*Corresponding authors.
[1]Code is available at: https://github.com/easm002/F2Net

range dependencies and the other capturing long-range semantic context. A Hybrid-Frequency Fusion (HFF) module integrates the outputs of these branches. To ensure effective multi-branch training, we introduce two complementary objectives: (1) Cross-Frequency Alignment Loss (CFAL) enforces semantic consistency between high- and low-frequency representations, facilitating coherent feature integration; and (2) Cross-Frequency Balance Loss (CFBL) regularizes the gradient magnitudes across branches, promoting balanced learning dynamics. Experiments on two UHR benchmarks - *DeepGlobe* [9] and *Inria Aerial* [21] - demonstrate F2Net's effectiveness.

In summary, our key contributions are as follows:

- A frequency-aware network - F2Net decomposes UHR RS images into high- and low-frequency components, enabling efficient and detail-preserving segmentation.
- We introduce two cross-frequency objectives for multi-branch learning on UHR imagery.
- We achieve the state-of-the-art performance on two UHR benchmarks—*DeepGlobe* and *Inria Aerial*.

## 2. Related Work

### 2.1. Segmentation

Semantic segmentation, a fundamental task in computer vision, aims to assign a categorical label to every pixel in an image. The field was revolutionized by Fully Convolutional Networks (FCN) [20], which first established the end-to-end, pixel-to-pixel training paradigm. Following this breakthrough, research diverged into several key strategies to address the inherent challenges of the task.

One primary line of work focused on recovering spatial details lost during encoding. U-Net [27] became the archetype for this encoder-decoder approach, using skip-connections to fuse deep, semantic features with shallow, fine-grained features. This concept was further refined by architectures like UNet++ [46], which introduced nested and dense skip pathways to reduce the semantic gap between encoder and decoder feature maps.

A second strategy prioritized the aggregation of multi-scale context, which FCNs struggled with due to their limited receptive fields. The DeepLab series [4] pioneered the use of atrous convolution to capture features at multiple scales without sacrificing spatial resolution. Concurrently, PSPNet [44] introduced a pyramid pooling module to effectively aggregate global context information.

A third major challenge was the poor localization of object boundaries. Early methods combined CNN outputs with post-processing steps like Conditional Random Fields (CRFs) to refine edges [3]. Later, attention mechanisms were integrated directly into the CNN architecture, such as in Attention U-Net [23], which uses attention gates to suppress irrelevant regions while highlighting salient features useful for the specific task.

More recently, the success of Vision Transformers (ViT) [10] has instigated a shift away from pure CNNs. Transformer-based models, such as SegFormer [38], leverage self-attention to effectively model long-range spatial dependencies. Further advancing the field, Mask2Former [7] proposed a novel mask classification paradigm, reformulating segmentation as a query-based set prediction problem.

### 2.2. Ultra High Resolution Segmentation

Standard segmentation methods struggle with UHR images due to memory constraints and structural complexity. Early solutions—downsampling or patch-wise processing—compromise global context or local detail [2, 45]. Recent advances adopt two strategies: Transformer-based models and multi-branch architectures.

Transformer-based approaches leverage the long-range dependency modeling capabilities of Transformers, often with modifications like limited positional encodings and lightweight decoders to manage memory overhead [38]. To further integrate local details, hybrid CNN-Transformer architectures have been proposed, combining the strengths of both backbones [43]. Some hybrid models introduce context-guided modules [14, 34] or patch-merging strategies [31] to enhance boundary consistency by aggregating information from neighboring patches. The multi-branch paradigm separates global and local processing, typically with a global branch handling downsampled inputs for context and a local branch focusing on high-resolution patches for detail [5, 12]. Extensions with multi-scale features, uncertainty estimation [25], and locality-aware designs help reduce boundary artifacts, but often increase latency due to overlapping patch inference [16].

Yet, these methods rely on spatial partitioning (e.g., cropping or patches), which can disrupt object integrity, our method enables simultaneous modeling of fine details and global context without introducing spatial artifacts.

### 2.3. Representation Learning in Frequency Domain

Representing images in the frequency domain is an increasingly effective technique in remote sensing, prized for its ability to capture both global context and fine-grained details [24, 29, 35, 40, 42]. Methods such as wavelet decomposition separate inputs into complementary low- and high-frequency components to produce coherent predictions [17]. Others employ adaptive frequency filtering to sharpen inter-class boundaries [29] or use Fast Fourier Convolution to add global context, thereby improving the detection of small objects [39]. Furthermore, transforms like DCT facilitate robust feature matching across varying conditions, such as illumination or seasonal shifts [42]. However, approaches that fuse multiple frequency components face challenges with fusion strategies and training stabil-

ity [6, 26, 28]. These issues are particularly critical yet largely unexplored in the ultra-high-resolution (UHR) remote sensing domain.

## 3. Methodology

This section presents our novel framework; specific implementations are detailed in Sec. 4. As shown in Figure 1, F2Net comprises high- and low-frequency branches, each specialized for distinct spectral representations to simplify learning and enhance task focus. A dynamic frequency decomposition module first splits the input into high- and low-frequency components. The high-frequency branch operates on full-resolution features to preserve fine structural details, while the low-frequency branch—capturing semantically rich but redundant signals—is downsampled to reduce computation without sacrificing contextual integrity. The low-frequency branch is further split into short- and long-range sub-branches to model local and global dependencies, respectively. A hybrid-frequency fusion (HFF) module then integrates features across branches. To facilitate multi-branch optimization, we introduce two objectives: (1) Cross-Frequency Alignment Loss (CFAL) promotes semantic consistency across frequencies; and (2) Cross-Frequency Balance Loss (CFBL) equalizes gradient flow, ensuring stable and balanced training.

### 3.1. Problem Formulation

Given a UHR remote sensing image, $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$, where $H$, $W$, and $C$ denote the height, width, and number of channels of the image, respectively, semantic segmentation aims to estimate a segmentation map $\mathbf{S}$, which is defined as:

$$\mathbf{S} = \{s^{hwl} \in \{0,1\} \mid 1 \le h \le H, 1 \le w \le W, 1 \le l \le L\}, \tag{1}$$

where $s^{hwl} = 1$ indicates that pixel at position $(h, w)$ belongs to semantic class $l$ and $s^{hwl} = 0$ otherwise. Here, $L$ denotes the total number of semantic classes. The predicted segmentation map generated by our method is denoted as $\hat{\mathbf{S}} = \{\hat{s}^{hwl}\} \in [0,1]^{H\times W\times L}$.

### 3.2. Adaptive Frequency Decomposition

We begin by applying a pointwise convolution to the input image $\mathbf{I}$ for cross-channel understanding as in [22], obtaining feature maps $\mathbf{X} \in \mathbb{R}^{H\times W\times D}$:

$$\mathbf{X} = \text{Conv}_{1\times 1}(\mathbf{I}). \tag{2}$$

A uniform kernel, typically, struggles to capture spatially varying high-frequency details, as signal frequency can change significantly across different regions of the image. To address this, we introduce a dynamic kernel formulation strategy based on channel-wise grouping. Specifically, the feature map $\mathbf{X}$ is uniformly split along the channel dimension into multiple ($N$) groups:

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_N], \quad \mathbf{X}_n \in \mathbb{R}^{H\times W\times \frac{D}{N}}. \tag{3}$$

For each group $\mathbf{X}_n$, we create spatially-adaptive low-pass filters through normalized convolution:

$$\mathbf{W}_n^{\text{LF}} = \text{Softmax}\left(\text{Conv}(\mathbf{X}_n)\right) \in \mathbb{R}^{H\times W\times k^2}, \tag{4}$$

where $k$ is the filter kernel size and the softmax operation apply along ($k^2$) dimension ensures non-negative weights that sum to one, enabling valid low-pass filtering behavior. The corresponding high-pass kernels are derived by subtracting from an identity kernel:

$$\mathbf{W}_n^{\text{HF}}[h,w,:] = \mathbf{1}_{k\times k} - \mathbf{W}_n^{\text{LF}}[h,w,:], \quad \forall (h,w). \tag{5}$$

These dynamic filters are then applied to their respective feature groups:

$$\mathbf{X}_n^{\text{LF}} = \mathbf{X}_n * \mathbf{W}_n^{\text{LF}}, \mathbf{X}_n^{\text{HF}} = \mathbf{X}_n * \mathbf{W}_n^{\text{HF}}, \tag{6}$$

where $*$ denotes depthwise convolution. Finally, the decomposed low- and high-frequency components are reconstructed via channel-wise concatenation.

$$\mathbf{X}^{\text{LF}} = [\mathbf{X}_1^{\text{LF}}, \ldots, \mathbf{X}_N^{\text{LF}}], \mathbf{X}^{\text{HF}} = [\mathbf{X}_1^{\text{HF}}, \ldots, \mathbf{X}_N^{\text{HF}}]. \tag{7}$$

The high-frequency branch adopts a multi-stage encoder to extract hierarchical representations from the high-frequency input $\mathbf{X}^{\text{HF}}$. Since its high resolution nature, State-Space Model (SSM) [19] based visual processing is adopted for the branch as SSM has shown significant improvement in the long-sequence modeling. The process begins with a convolutional embedding layer, projecting the input into an initial feature space:

$$\mathbf{Z}^{(0)} = \text{Conv2D}(\mathbf{X}^{\text{HF}}) \in \mathbb{R}^{H_0\times W_0\times C_0} \tag{8}$$

Feature extraction is then performed through multiple stages, each consisting of $L$ consecutive Visual State Space (VSS) blocks. Within each block, the following operations are sequentially applied: 1) layer normalization is used to normalize the input features:

$$\mathbf{Z}_{\text{norm}}^{(l)} = \text{LayerNorm}\left(\mathbf{Z}^{(l-1)}\right). \tag{9}$$

Then, an SS2D [19] captures spatial dependencies, and its output is combined with the residual input:

$$\mathbf{Z}_{\text{SS2D}}^{(l)} = \mathbf{Z}^{(l-1)} + \text{SS2D}\left(\mathbf{Z}_{\text{norm}}^{(l)}\right). \tag{10}$$

This is followed by a second normalization and passed through a feed-forward network (FFN). The final output of the block is computed as:

$$\mathbf{Z}^{(l)} = \mathbf{Z}_{\text{SS2D}}^{(l)} + \text{FFN}\left(\text{LayerNorm}\left(\mathbf{Z}_{\text{SS2D}}^{(l)}\right)\right). \tag{11}$$

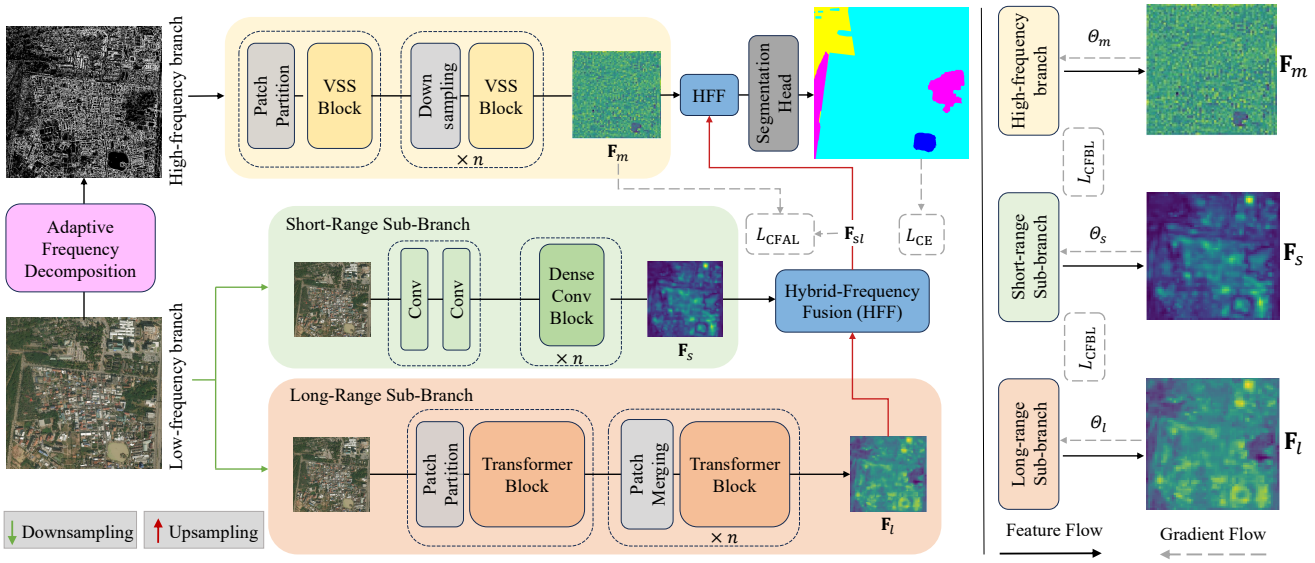



Figure 1. Overview of F2Net's multi-branch design. An adaptive frequency decomposition module splits the input into high- and low-frequency components. The high-frequency branch preserves details, while the low-frequency branch (with downsampling) includes short- and long-range sub-branches. A hybrid-frequency fusion (HFF) module integrates all features. Two auxiliary losses—Cross-Frequency Alignment Loss (CFAL) and Cross-Frequency Balance Loss (CFBL)—promote semantic consistency and balanced optimization.

Downsampling is applied between consecutive stages to progressively enlarge the receptive field while maintaining computational efficiency. The final output of the high-frequency branch is denoted $\mathbf{F}_m \in \mathbb{R}^{H_m \times W_m \times C_m}$.

### 3.3. Low-Frequency Branch

For comprehensive low-frequency semantic modeling, the downsampled low-frequency component $\mathbf{X}^{\text{LF}}$ is processed through two complementary subbranches. The short-range subbranch, built upon a dense convolutional block, producing an output $\mathbf{F}_s \in \mathbb{R}^{H_s \times W_s \times C_s}$. To address the limited receptive field of CNNs, we introduce a long-range subbranch based on TinyViT [36], which processes the input $\mathbf{X}^{\text{LF}}$ to model long-range dependencies and generate global feature representations $\mathbf{F}_l \in \mathbb{R}^{H_l \times W_l \times C_l}$.

### 3.4. Hybrid-Frequency Fusion

Heterogeneous features from different branches may contain redundant or misaligned channels, which can hinder effective integration. To address this, we introduce a Hybrid-Frequency Fusion (HFF) module that performs hierarchical feature aggregation. Specifically, HFF first integrates the outputs of the two low-frequency sub-branches to capture both short- and long-range context, and then fuses this representation with the high-frequency branch output to form the final unified feature.

In detail, we first apply a channel-wise attention mechanism to each low-frequency sub-branch, enhancing informative channels while suppressing less relevant ones prior to fusion. Given $\mathbf{F}_s$ and $\mathbf{F}_l$, HFF evaluates the corresponding channel attentions $\mathbf{A}_s$ and $\mathbf{A}_l$:

$$\begin{aligned} \mathbf{A}_s &= \sigma(\text{MLP}(\text{Pool}(\mathbf{F}_s))) \in \mathbb{R}^{C_s}, \\ \mathbf{A}_l &= \sigma(\text{MLP}(\text{Pool}(\mathbf{F}_l))) \in \mathbb{R}^{C_l}, \end{aligned} \tag{12}$$

where $\sigma$ represents a sigmoid activation function, MLP denotes a multi-layer perceptron, and Pool indicates spatial average pooling. Next, we model cross-branch relationships:

$$\mathbf{M} = \sigma(\mathbf{A}_s \mathbf{A}_l^{\text{T}}). \tag{13}$$

The cross-branch attention matrix $\mathbf{M} \in \mathbb{R}^{C_s \times C_l}$ is incorporated with the original channel attention to enhance feature interaction, allowing for more effective information exchange between different branches. It is achieved via two separate MLPs for dimension matching purposes:

$$\tilde{\mathbf{A}}_s = \sigma(\text{MLP}_s(\mathbf{M}) + \mathbf{A}_s), \tilde{\mathbf{A}}_l = \sigma(\text{MLP}(\mathbf{M}) + \mathbf{A}_l). \tag{14}$$

We then apply these refined attentions to their corresponding branches and fuse them together:

$$\mathbf{F}_{\text{sl}} = \text{Conv}(\tilde{\mathbf{A}}_s \cdot \mathbf{F}_s) + \text{Conv}(\tilde{\mathbf{A}}_l \cdot \mathbf{F}_l), \tag{15}$$

where two separate $1 \times 1$ convolution operations project features into a unified channel dimension to facilitate proper feature fusion.

The fused low-frequency feature $\mathbf{F}_{\text{sl}}$ is subsequently combined with the high-frequency output $\mathbf{F}_m$ via a second HFF module, yielding the final integrated representation. This comprehensive feature is then forwarded to the segmentation head to produce the final prediction.

### 3.5. Cross-Frequency Loss & Optimization

**Cross-Frequency Alignment Loss**. In F2Net, each frequency branch focuses on a distinct spectral component. In specific, the high-frequency branch and low-frequency branch are inherently asymmetric in both representation and semantic abstraction. Although these branches target complementary aspects of the image, they often encode different representations of the same semantic content. Without explicit alignment, this discrepancy may lead to semantic inconsistency during feature fusion and segmentation. To enforce coherent semantic understanding across high- and low-frequency branches, we introduce a cross-frequency alignment loss (CFAL). It encourages the branch outputs to converge at the semantic level, ensuring that the same object class is consistently represented across branches regardless of frequency bias. We implement CFAL using a symmetric Kullback–Leibler (KL) divergence, defined as:

$$\mathcal{L}_{\text{CFAL}} = \frac{1}{2}\left[D_{\text{KL}}(\mathbf{F}_{\text{sl}} \parallel \mathbf{F}_{\text{m}}) + D_{\text{KL}}(\mathbf{F}_{\text{m}} \parallel \mathbf{F}_{\text{sl}})\right], \quad (16)$$

**Cross-Frequency Balance Loss**. Apart from the alignment issue, the asymmetry can lead to structurally imbalanced gradient magnitudes across branches during backpropagation [6, 28]. Such imbalance can cause the optimization process to disproportionately favor one branch over the other, undermining the collaborative learning objective and resulting in inefficient frequency feature fusion. To address this, we propose a cross-frequency balance loss (CFBL) that explicitly regularizes the gradient norms of each frequency branch, ensuring that no single branch dominates the learning dynamics. Formally, the CFBL is defined as:

$$\mathcal{L}_{\text{CFBL}} = \sum_{\Theta} |G_{\Theta} - \bar{G}|, \quad (17)$$

where $G_i(t) = ||\nabla_{\Theta}\mathcal{L}_{\text{CE}}||_2$ denotes the gradient magnitude with respect to a conventional segmentation cross-entropy loss $\mathcal{L}_{\text{CE}}$ for the parameters $\Theta$ of a particular branch, and $\bar{G}$ is the mean gradient magnitude across all branches.

**Optimization**. The overall loss $\mathcal{L}$ used for optimization combines the proposed cross-frequency losses: CFAL and CFBL, together with the segmentation loss $\mathcal{L}_{\text{CE}}$. It is formulated as a weighted sum:

$$\mathcal{L} = \lambda_1\mathcal{L}_{\text{CFAL}} + \lambda_2\mathcal{L}_{\text{CFBL}} + \lambda_3\mathcal{L}_{\text{CE}}, \quad (18)$$

where $\lambda_1$, $\lambda_2$, and $\lambda_3$ are hyperparameters.

## 4. Experiment

### 4.1. Implementation & Evaluation Protocol

**Dataset**. We evaluate our method on two benchmarks. *DeepGlobe* [9] contains 803 satellite images ($2448 \times 2448$), split into 455/207/142 for training/validation/testing, with annotations for seven land-cover classes (urban, agriculture, rangeland, forest, water, barren, unknown), where the unknown class is excluded [12]. *Inria Aerial* [21] includes 180 aerial images ($5000 \times 5000$) annotated with binary building masks, following the split 126/27/27 for training/validation/testing as in [16].

**Implementation Details.** The short-range subbranch is implemented using DeepLabv3 [4] with a customized ResNet-18 backbone. The long-range subbranch adopts a 6-layer ViT-tiny architecture [10]. All backbones are initialized with weights pre-trained on ImageNet-1K. The Mamba backbone is built upon VMamba-Tiny-M2 [19], featuring a hierarchical structure of four stages with depths [2, 2, 4, 2], and a base channel dimension 64. We set the channel grouping factor $N$ in Eq. 3 to 8. We optimize all parameters using SGD with momentum 0.9, an initial learning rate of $1 \times 10^{-3}$, and polynomial decay (power 0.9). Loss weights are set as $\lambda_1 = \lambda_2 = 0.1$, $\lambda_3 = 1$. Training runs for 40k iterations on Inria Aerial and 80k on DeepGlobe, with batch size 8 on a DGX-1 workstation (Tesla V100 GPUs).

**Baselines & metrics.** We compare F2Net with state-of-the-art methods in two groups: *(i) Conventional segmentation*—FCN-8s [20], UNet [27], DeepLabv3+ [4], STDC [11], SegFormer [38], Mask2Former [7], and PointRend [15]; *(ii) UHR-specific*—CascadePSP [8], PPN [37], MagNet [13], GLNet [5], FCtL [16], ISDNet [12], SGNet [34], GPWFormer [14], RUE [25], and BPT [31]. We report accuracy via F1 and mIoU, and efficiency via FPS and memory. All FPS are measured on a single NVIDIA Tesla V100 for fair comparison.

### 4.2. Comparison with State-of-the-Arts

**Quantitative comparison**. Quantitative results are reported in Table 1 and Table 2 for DeepGlobe and Inria Aerial, respectively. We compare conventional models adapted by cropping or downsampling and UHR-specific methods. On DeepGlobe, downsampling consistently improves throughput over cropping and often improves accuracy for UNet and DeepLabv3+, while FCN-8s is a counterexample where cropping attains a slightly higher mIoU (71.80 vs 68.80). UHR-specific methods are stronger overall: many exceed 70% mIoU, and our F2Net is the only model above 80% on DeepGlobe, reaching 80.22 mIoU and 87.09 F1 at 24.30 FPS. This yields gains of +3.62 mIoU over BPT (76.60) and +4.78 over SGNet (75.44), while remaining the third-fastest among methods with reported FPS (behind RUE at 44.38 FPS and ISDNet at 27.70 FPS). On

Inria Aerial, F2Net attains 83.39 mIoU and 91.19 F1, improving the previous best mIoU (RUE, 79.00) by +4.39. In terms of memory, F2Net uses 2,767 MB on DeepGlobe and 5,534 MB on Inria Aerial, representing a deliberate accuracy–efficiency trade-off at ultra-high resolutions. Overall, F2Net delivers the highest accuracy on both datasets while maintaining practical inference speed.

| Method | mIoU↑ | F1↑ | FPS↑ | Mem. (MB)↓ |
|---|---|---|---|---|
| **Conventional Segmentation** | | | | |
| *Cropping* | | | | |
| UNet [27] | 47.20 | 64.2 | 1.30 | 1,730 |
| FCN-8s [20] | 71.80 | 82.6 | 4.55 | **970** |
| DeepLabv3+ [4] | 64.10 | 78.3 | 1.50 | 1,530 |
| SegFormer [38] | 63.40 | 73.5 | 1.20 | 2,200 |
| Mask2Former [7] | 63.90 | 74.1 | 0.80 | 4,500 |
| *Downsampling* | | | | |
| UNet [27] | 64.10 | 78.3 | 3.60 | 7,600 |
| FCN-8s [20] | 68.80 | 79.8 | 7.91 | 1,984 |
| DeepLabv3+ [4] | 70.10 | 82.5 | 4.50 | 3,210 |
| STDC [11] | **71.20** | **83.1** | **13.80** | 2,570 |
| **UHR Segmentation** | | | | |
| CascadePSP [8] | 68.50 | 79.7 | 0.11 | 3,236 |
| PPN [37] | 71.90 | – | 12.90 | 1,193 |
| PointRend [15] | 72.00 | 83.9 | 6.30 | 1,600 |
| MagNet [13] | 73.20 | 84.6 | 0.82 | 1,565 |
| GLNet [5] | 71.60 | 83.2 | 0.17 | 1,973 |
| FCtL [16] | 73.50 | 83.8 | 0.13 | 4,450 |
| ISDNet [12] | 73.30 | 85.18 | 27.70 | 1,948 |
| SGNet [34] | 75.44 | – | 0.66 | 2,187 |
| GPWFormer [14] | 75.80 | 85.4 | – | 2,380 |
| RUE [25] | 75.10 | – | **44.38** | **1,638** |
| BPT [31] | 76.60 | 85.7 | – | 2,074 |
| Ours | **80.22** | **87.09** | 24.30 | 2,767 |

Table 1. Performance comparison on **DeepGlobe**. Quantitative results of conventional and UHR segmentation methods, showing mIoU, F1, FPS, and memory usage.

| Method | mIoU↑ | F1↑ | FPS↑ | Mem. (MB)↓ |
|---|---|---|---|---|
| **Conventional Segmentation** | | | | |
| FCN-8s [20] | 69.10 | 81.70 | 1.90 | 4050 |
| DeepLabv3+ [4] | 55.90 | – | 1.67 | 5122 |
| STDC [11] | 72.44 | – | 4.97 | 7410 |
| **UHR Segmentation** | | | | |
| GLNet [5] | 71.20 | – | 0.05 | **2663** |
| CascadePSP [8] | 69.40 | 81.80 | 0.03 | 3236 |
| FCtL [16] | 73.70 | 84.10 | 0.04 | 4330 |
| ISDNet [12] | 74.23 | 86.35 | 6.90 | 4680 |
| GPWFormer [14] | 76.50 | – | – | 4710 |
| BPT [31] | 77.10 | 86.30 | – | 4450 |
| RUE [25] | 79.00 | – | **10.89** | 3574 |
| Ours | **83.39** | **91.19** | 5.50 | 5534 |

Table 2. **Inria Aerial** results for conventional and UHR methods: mIoU, F1, FPS, and memory.

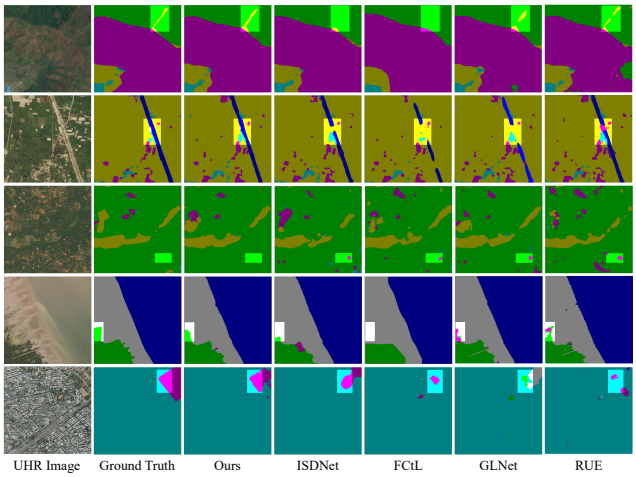



Figure 2. Qualitative examples from the DeepGlobe dataset. Each color denotes a distinct semantic region: cyan for urban, yellow for agriculture, purple for rangeland, green for forest, blue for water, white for barren.

**Qualitative Analysis.** Figures 2 and 3 present qualitative comparisons with key regions highlighted. These results demonstrate our model's capability across diverse scenarios: DeepGlobe examples illustrate effective exploitation of large-scale semantic context for natural landscape disambiguation, while Inria examples showcase accurate capture of high-frequency structural details in dense urban settings.

In DeepGlobe (Fig. 2), our model exhibits strong long-range contextual modeling. In the first row, it accurately segments a narrow agricultural strip embedded within forest, maintaining feature continuity and topological integrity—whereas competing methods (*e.g.*, FCtL, GLNet) produce fragmented or missed detections. Similar behavior is observed in the second row, where our model correctly traces a continuous watercourse, while other SOTAs either fragment this linear structure (ISDNet, FCtL) or introduce substantial false positives along riverbanks (RUE). Our model also demonstrates robust contextual reasoning that mitigates common misclassifications across scales. In the third row, competing methods misclassify large bare patches within forest as rangeland due to insufficient contextual information. Conversely, in the fourth row, they fail to recognize a small forest patch, incorrectly labeling it as surrounding barren land. Our model correctly classifies both cases, evidencing effective contextual awareness for both large regions and small isolated objects. The fifth

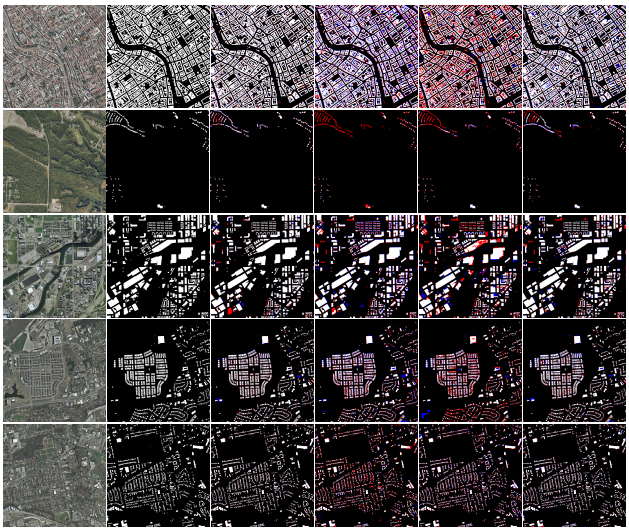



Figure 3. Qualitative examples from the Inria Aerial dataset. White and black pixels denote building and non-building regions, respectively. For error visualization, red pixels highlight false negatives, and blue pixels indicate false positives in building detection.

| High-frequency | Short-range | Long-range | mIoU↑ |
|---|---|---|---|
| ✓ | | | 72.5 |
| | ✓ | | 71.3 |
| | | ✓ | 71.9 |
| | ✓ | ✓ | 73.7 |
| ✓ | | ✓ | 76.5 |
| ✓ | ✓ | | <u>76.7</u> |
| ✓ | ✓ | ✓ | **80.22** |

Table 3. Ablation study on the impact of different branch combinations on **DeepGlobe**. Best result in bold, second-best underlined.

| Hi-frequency | Short | Long | mIoU↑ | Mem. (MB)↓ |
|---|---|---|---|---|
| Full | 1/4 | 1/4 | 80.22 | 2,767 |
| Full | 1/8 | 1/8 | 76.5 | 2,133 |
| 1/2 | 1/4 | 1/4 | 75.1 | 2,789 |
| 1/4 | 1/4 | 1/4 | 72.8 | 2,283 |
| 1/4 | 1/2 | 1/2 | — | — |

Table 4. Ablation study on the input resolution of different branches on DeepGlobe. "Short" and "Long" denote the short- and long-range branches.

row further validates this capability, where our method produces the most accurate boundary delineation for a rangeland patch within an urban block.

In the Inria Aerial dataset (Fig. 3), where false negatives (red) and false positives (blue) visualize segmentation errors, our model achieves superior spatial precision. The second row illustrates this capability: our model correctly detects the building in the top-left region (missed by ISDNet) while producing accurate boundaries for the complex building cluster in the top-right, substantially reducing classification errors compared to FCtL and RUE. This precision in resolving high-frequency details is consistent across examples. In the dense scenes of rows three and four, our segmentation exhibits the lowest error rates. In row five, containing numerous discrete buildings, our method accurately captures fine structures that baselines miss. Detailed class-wise gains demonstrating F2Net's contextual reasoning are in the supplementary material.

### 4.3. Ablation Studies

**Effectiveness of frequency-aware branches.** We investigate the contribution of each branch in F2Net on DeepGlobe (Table 3). The high-frequency branch alone achieves 72.5% mIoU, demonstrating that full-resolution frequency cues effectively capture boundary details while preserving global dependencies via the Mamba backbone. In contrast, the short-range CNN and long-range Transformer branches, both operating on downsampled inputs, yield 71.3% and 71.9%, respectively, reflecting their limited ability to recover fine boundaries after resolution reduction. When paired with the high-frequency branch, both short- and long-range branches lead to clear gains (76.7% and 76.5%), showing that local texture and global semantic reasoning complement frequency features in different ways. Finally, the three-branch configuration reaches 80.22% mIoU, exceeding the best dual-branch combination by 3.5 points and the average single-branch result by over 8 points. This consistent improvement verifies the strong synergistic effect among frequency, local, and global pathways, suggesting that integrating full-resolution high-frequency cues with multi-scale spatial reasoning is critical for accurate and robust UHR segmentation.

**Impact of input resolution.** Table 4 evaluates the influence of input resolution on each branch. The high-frequency branch is highly resolution-sensitive: reducing its input to 1/2 and 1/4 of the original size leads to a substantial mIoU drop of 5.1% and 7.4%, respectively, confirming that full-resolution inputs are essential for preserving boundary and texture details. In contrast, the short- and long-range branches exhibit better robustness to downsampling. Lowering their resolution from 1/4 to 1/8 reduces memory by 23% with a moderate 3.72% performance loss, indicating that global and local semantic reasoning can be efficiently modeled at coarser scales. Given the 16GB GPU memory constraint during *training*, the resolution of low-frequency branches is limited to half of the original input. Overall, these findings validate the core princi-

| $\mathcal{L}_{CE}$ | $\mathcal{L}_{CFBL}$ | $\mathcal{L}_{CFAL}$ | mIoU↑ |
|---|---|---|---|
| ✓ | | | 77.35 |
| ✓ | ✓ | | 78.80 |
| ✓ | | ✓ | 79.20 |
| ✓ | ✓ | ✓ | **80.22** |

Table 5. Ablation study on different loss functions.

| Fusion Method | mIoU↑ | GFLOPs↓ |
|---|---|---|
| Concat | 76.10 | 48.7 |
| Add | 74.60 | 45.2 |
| ASPP | 77.30 | 61.5 |
| Cross-attention | 72.40 | 93.8 |
| Our HFF | **80.22** | 60.8 |

Table 6. Ablation study on different feature fusion strategies. "Our HFF" denotes our proposed fusion method. Best mIoU is in bold.

ple of our frequency-aware design: full-resolution processing is indispensable for high-frequency details, while low-frequency semantics can be effectively captured at reduced resolutions—achieving a balanced trade-off between accuracy and computational cost.

**Effectiveness of loss functions**. Table 5 analyzes the impact of each loss component in our framework. Using only the standard Cross-Entropy (CE) loss yields 77.35% mIoU, serving as the baseline. Introducing the Cross-Frequency Balance Loss (CFBL) improves performance to 78.80% by stabilizing gradient magnitudes across heterogeneous branches, thus facilitating balanced optimization between high- and low-frequency pathways. In parallel, adding the Cross-Frequency Alignment Loss (CFAL) alone increases mIoU to 79.20%, confirming its effectiveness in aligning frequency-domain features and enhancing inter-branch semantic consistency. When both CFBL and CFAL are jointly applied with CE, F2Net attains the best performance of 80.22%, outperforming the CE-only baseline by 2.87%. This consistent improvement verifies that CFBL and CFAL act in complementary manners—one enforcing training stability, the other ensuring semantic coherence—together forming a cohesive objective that maximizes the synergy among frequency-aware representations. Further empirical analysis detailing the gradient norm ratios and demonstrating the stabilization effect of CFBL is provided in the supplementary material.

**Hybrid-Frequency Fusion Analysis.** We further evaluate the proposed Hybrid-Frequency Fusion (HFF) module against representative fusion strategies summarized in Table 6. Simple addition yields the lowest computational cost (45.2 GFLOPs) but only achieves 74.6% mIoU, indicating limited capability in integrating heterogeneous frequency cues. Feature concatenation improves performance to 76.1% by retaining branch-specific information, yet its fusion remains shallow and lacks adaptive weighting. ASPP attains a moderate 77.3% mIoU by enlarging the receptive field through multi-scale spatial context, but it ignores frequency interactions and increases computation (61.5 GFLOPs). Cross-attention, though theoretically capable of modeling inter-branch dependencies, performs the worst (72.4%) due to excessive computation (93.8 GFLOPs) and overfitting to redundant correlations. Our HFF achieves the highest mIoU of 80.22% with moderate complexity (60.8 GFLOPs). This improvement stems from its frequency-wise gating mechanism, which adaptively aligns multi-frequency features rather than simply merging them, enabling more effective cross-frequency interaction and balanced representation learning.

| Method | mIoU↑ | FPS↑ |
|---|---|---|
| Raw Input | 74.20 | 11.3 |
| Gaussian filter | 76.00 | 12.1 |
| Laplacian Pyramid | 78.10 | 9.8 |
| AFD (Ours) | **80.22** | 15.4 |

Table 7. Ablation study on frequency decomposition strategies.

**Frequency Decomposition Analysis.** We evaluate frequency decomposition strategies applied to the high-frequency branch summarized in Table 7. Directly processing raw full-resolution inputs yields the lowest mIoU of 74.2% due to the lack of explicit frequency separation. A Gaussian filter improves performance to 76.0% by suppressing noise but offers limited adaptability. The Laplacian Pyramid achieves higher accuracy (78.1%) by capturing multi-scale details, yet its fixed decomposition introduces additional computation and limits real-time efficiency (9.8 FPS). In contrast, our Adaptive Frequency Decomposition (AFD) attains the best balance between accuracy and efficiency, reaching 80.22% mIoU with 15.4 FPS. This gain arises from dynamic content-aware filtering, which allocates frequency emphasis based on spatial context, preserving fine structures while accelerating inference.

## 5. Conclusion

We present F2Net, a frequency-aware framework for semantic segmentation of UHR remote sensing images. Decomposing inputs into high- and low-frequency components allows F2Net to process structural details and semantics in parallel via specialized branches. The Hybrid-Frequency Fusion module integrates multi-scale features, while cross-frequency objectives stabilize training. Experiments on two widely-used benchmarks - *DeepGlobe* and *Inria Aerial* - show state-of-the-art results.

## Acknowledgments

This work was partially supported by the ECU Early-Mid Career Researcher (EMCR) Grant Scheme.

## References

[1] S. Ascher and E. Pincus. *The Filmmaker's Handbook: A Comprehensive Guide for the Digital Age*. Plume, 1999. 1

[2] Liang-Chieh Chen, Yi Yang, Jiang Wang, Wei Xu, and Alan L. Yuille. Attention to scale: Scale-aware semantic image segmentation. In *CVPR*, 2016. 1, 2

[3] Liang-Chieh Chen, George Papandreou, Iasonas Kokkinos, Kevin Murphy, and Alan L. Yuille. Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2018. 2

[4] Liang-Chieh Chen, Yukun Zhu, George Papandreou, Florian Schroff, and Hartwig Adam. Encoder-decoder with atrous separable convolution for semantic image segmentation. In *ECCV*, 2018. 2, 5, 6

[5] Wuyang Chen, Ziyu Jiang, Zhangyang Wang, Kexin Cui, and Xiaoning Qian. Collaborative global-local networks for memory-efficient segmentation of ultra-high resolution images. In *CVPR*, 2019. 1, 2, 5, 6

[6] Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *ICML*, 2018. 3, 5

[7] Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *CVPR*, 2022. 2, 5, 6

[8] Ho Kei Cheng, Jihoon Chung, Yu-Wing Tai, and Chi-Keung Tang. Cascadepsp: Toward class-agnostic and very high-resolution segmentation via global and local refinement. In *CVPR*, 2020. 5, 6

[9] Ilke Demir, Krzysztof Koperski, David Lindenbaum, Guan Pang, Jing Huang, Saikat Basu, Forest Hughes, Devis Tuia, and Ramesh Raskar. Deepglobe 2018: A challenge to parse the earth through satellite images. In *CVPRW*, 2018. 2, 5

[10] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021. 2, 5

[11] Mingyuan Fan, Shenqi Lai, Junshi Huang, Xiaoming Wei, Zhenhua Chai, Junfeng Luo, and Xiaolin Wei. Rethinking bisenet for real-time semantic segmentation. In *CVPR*, 2021. 5, 6

[12] Shaohua Guo, Liang Liu, Zhenye Gan, Yabiao Wang, Wuhao Zhang, Chengjie Wang, Guannan Jiang, Wei Zhang, Ran Yi, Lizhuang Ma, and Ke Xu. Isdnet: Integrating shallow and deep networks for efficient ultra-high resolution segmentation. In *CVPR*, 2022. 1, 2, 5, 6

[13] Chuong Huynh, Anh Tuan Tran, Khoa Luu, and Minh Hoai. Progressive semantic segmentation. In *CVPR*, 2021. 5, 6

[14] Deyi Ji, Feng Zhao, and Hongtao Lu. Guided patch-grouping wavelet transformer with spatial congruence for ultra-high resolution segmentation. In *IJCAI*, 2023. 2, 5, 6

[15] Alexander Kirillov, Yuxin Wu, Kaiming He, and Ross Girshick. Pointrend: Image segmentation as rendering. In *CVPR*, 2020. 5, 6

[16] Qi Li, Weixiang Yang, Wenxi Liu, Yuanlong Yu, and Shengfeng He. From contexts to locality: Ultra-high resolution image segmentation via locality-aware contextual correlation. In *ICCV*, 2021. 2, 5, 6

[17] Yifan Li, Ziqian Liu, Junli Yang, and Haopeng Zhang. Wavelet transform feature enhancement for semantic segmentation of remote sensing images. *Remote Sensing*, 2023. 2

[18] Ziming Li, Bin Chen, Shengbiao Wu, Mo Su, Jing M. Chen, and Bing Xu. Deep learning for urban land use category classification: A review and experimental assessment. *Remote Sensing of Environment*, 2024. 1

[19] Yue Liu, Yunjie Tian, Yuzhong Zhao, Hongtian Yu, Lingxi Xie, Yaowei Wang, Qixiang Ye, Jianbin Jiao, and Yunfan Liu. Vmamba: Visual state space model. In *NeurIPS*, 2024. 3, 5

[20] Jonathan Long, Evan Shelhamer, and Trevor Darrell. Fully convolutional networks for semantic segmentation. In *CVPR*, 2015. 2, 5, 6

[21] Emmanuel Maggiori, Yuliya Tarabalka, Guillaume Charpiat, and Pierre Alliez. Can semantic labeling methods generalize to any city? the inria aerial image labeling benchmark. In *IGARSS*, 2017. 2, 5

[22] Salma Abdel Magid, Yulun Zhang, Donglai Wei, Won-Dong Jang, Zudi Lin, Yun Fu, and Hanspeter Pfister. Dynamic high-pass filtering and multi-spectral attention for image super-resolution. In *ICCV*, 2021. 3

[23] Ozan Oktay, Jo Schlemper, Loic Le Folgoc, Matthew Lee, Mattias Heinrich, Kazunari Misawa, Kensaku Mori, Steven McDonagh, Nils Y Hammerla, Bernhard Kainz, et al. Attention u-net: Learning where to look for the pancreas. In *Medical Imaging with Deep Learning*, 2018. 2

[24] Botong Ou, Gang Shao, Baijian Yang, and Songlin Fei. Focalsr: Revisiting image super-resolution transformers with fourier-transform cross attention layers for remote sensing image enhancement. *Geomatica*, 2025. 2

[25] Rong Qin, Xingyu Liu, Jinglei Shi, Liang Lin, and Jufeng Yang. Boosting the dual-stream architecture in ultra-high resolution segmentation with resolution-biased uncertainty estimation. In *CVPR*, 2025. 2, 5, 6

[26] Nasim Rahaman, Aristide Baratin, Devansh Arpit, Felix Draxler, Min Lin, Fred Hamprecht, Yoshua Bengio, and Aaron Courville. On the spectral bias of neural networks. In *ICML*, 2019. 3

[27] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *MICCAI*, 2015. 2, 5, 6

[28] Ozan Sener and Vladlen Koltun. Multi-task learning as multi-objective optimization. In *NeurIPS*, 2018. 3, 5

[29] Binbin Song, Songhan Min, Hui Yang, Yongchuang Wu, and Biao Wang. A fourier frequency domain convolutional neu-

ral network for remote sensing crop classification considering global consistency and edge specificity. *Remote Sensing*, 2023. 2

[30] Robin Strudel, Ricardo Garcia, Ivan Laptev, and Cordelia Schmid. Segmenter: Transformer for semantic segmentation. In *ICCV*, 2021. 1

[31] Haopeng Sun, Yingwei Zhang, Lumin Xu, Sheng Jin, and Yiqiang Chen. Ultra-high resolution segmentation via boundary-enhanced patch-merging transformer. In *AAAI*, 2025. 2, 5, 6

[32] Yuxi Sun, Shanshan Feng, Xutao Li, Yunming Ye, Jian Kang, and Xu Huang. Visual grounding in remote sensing images. In *ACMMM*, 2022. 1

[33] Di Wang, Jing Zhang, Bo Du, Minqiang Xu, Lin Liu, Dacheng Tao, and Liangpei Zhang. Samrs: Scaling-up remote sensing segmentation dataset with segment anything model. In *NeurIPS*, 2023. 1

[34] Sai Wang, Yutian Lin, Yu Wu, and Bo Du. Toward real ultra image segmentation: Leveraging surrounding context to cultivate general segmentation model. In *NeurIPS*, 2024. 1, 2, 5, 6

[35] Penghui Wen, Mengwei He, Patrick Filippi, Na Zhao, Feng Zhang, Thomas Francis Bishop, Zhiyong Wang, and Kun Hu. DuoCast: Duo-probabilistic diffusion for precipitation nowcasting. In *AAAI*, 2026. 2

[36] Kan Wu, Jinnian Zhang, Houwen Peng, Mengchen Liu, Bin Xiao, Jianlong Fu, and Lu Yuan. Tinyvit: Fast pretraining distillation for small vision transformers. In *ECCV*, 2022. 4

[37] Tong Wu, Zhenzhen Lei, Bingqian Lin, Cuihua Li, Yanyun Qu, and Yuan Xie. Patch proposal network for fast semantic segmentation of high-resolution images. In *AAAI*, 2020. 5, 6

[38] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. Segformer: Simple and efficient design for semantic segmentation with transformers. In *NeurIPS*, 2021. 2, 5, 6

[39] Hui Yang, Caili Zhou, Xiaoyu Xing, Yongchuang Wu, and Yanlan Wu. A high-resolution remote sensing road extraction method based on the coupling of global spatial features and fourier domain features. *Remote Sensing*, 2024. 2

[40] Yunsong Yang, Genji Yuan, and Jinjiang Li. Sffnet: A wavelet-based spatial and frequency domain fusion network for remote sensing segmentation. *IEEE Transactions on Geoscience and Remote Sensing*, 2024. 2

[41] Xin Yu, Peng Dai, Wenbo Li, Lan Ma, Jiajun Shen, Jia Li, and Xiaojuan Qi. Towards efficient and scale-robust ultra-high-definition image demoiréing. In *ECCV*, 2022. 1

[42] Qinyan Zeng, Bin Hui, Zhaoji Liu, Zheng Xu, and Miao He. A method combining discrete cosine transform with attention for multi-temporal remote sensing image matching. *Sensors*, 2025. 2

[43] Cheng Zhang, Wanshou Jiang, Yuan Zhang, Wei Wang, Qing Zhao, and Chenjie Wang. Transformer and cnn hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery. *IEEE Transactions on Geoscience and Remote Sensing*, 2022. 2

[44] Hengshuang Zhao, Jianping Shi, Xiaojuan Qi, Xiaogang Wang, and Jiaya Jia. Pyramid scene parsing network. In *CVPR*, 2017. 2

[45] Hengshuang Zhao, Xiaojuan Qi, Xiaoyong Shen, Jianping Shi, and Jiaya Jia. Icnet for real-time semantic segmentation on high-resolution images. In *ECCV*, 2018. 1, 2

[46] Zongwei Zhou, Md Mahfuzur Rahman Siddiquee, Nima Tajbakhsh, and Jianming Liang. Unet++: A nested u-net architecture for medical image segmentation. In *DLMIA*, 2018. 2